

Leo FLACK

+33 7 85 71 47 14 | lh31@cam.ac.uk | leo.flack@polytechnique.edu
leof333.github.io | github.com/leoF333 | linkedin.com/in/leoflack

EDUCATION

University of Cambridge <i>Part III of the Mathematical Tripos (MASt in Mathematics)</i> • Modules in <i>Probability, Advanced Statistical Physics, Information Theory, Quantum Information & Computation.</i>	Oct 2026 – Jun 2027 Cambridge, UK
Ecole Polytechnique <i>Ingenieur Polytechnicien – Mathematics & Physics</i> • GPA: 3.97/4.0 . Selected coursework: <i>Martingales & Markov Chains, Optimisation & Control, Statistical Physics & Complex Systems, Advanced Quantum Mechanics.</i>	Sep 2023 – Aug 2026 Palaiseau, France
Lycee Louis-le-Grand <i>Classes Préparatoires aux Grandes Ecoles (CPGE) – Mathematics & Physics (MP*)</i>	Sep 2021 – Jun 2023 Paris, France

PUBLICATIONS

Carer, C., Verdier, S., Flack, L. , Scemama, A. & Petronio, F. “Global plasma modeling of an air-breathing gridded ion thruster for very low Earth orbit applications.” <i>Physics of Plasmas</i> 33 (8), 2026. DOI: 10.1063/5.0327675	
In preparation: <i>Finite Is Different: Birth of Sequential Recall in Delayed Hopfield Networks.</i>	

HONORS & AWARDS

National University Entrance Exams (2023): ranked 2nd / 5,000+ (Mines Paris), 6th (Ecole Polytechnique), 9th (ENS Ulm)	
Saven European Scholarship (2026): £25,000 award (Oxford physics, declined for Cambridge math)	
Carnot Scholarship (2026): merit-based award, 1 recipient in cohort , Ecole Polytechnique	
Military Decorations (2024): NATO Vigilance Medal; French Bronze Medal of National Defence for exceptional operational service	

RESEARCH EXPERIENCE

Finite-Size Bifurcations in Delayed Hopfield Networks – UChicago <i>Research Intern – Advisor: Prof. Vincenzo Vitelli</i> • Demonstrated that finite delayed non-reciprocal Hopfield networks differ qualitatively from the thermodynamic limit: each stored memory loses stability at a sample-specific saddle-node threshold. • Built a Python pipeline for numerical continuation, DDE integration, spectral stability and Lyapunov analysis; showed that the near-Gaussian threshold bulk destroys static recall, while its extreme predicts the onset of sequential recall.	Mar 2026 – Aug 2026 Chicago, IL, USA
Statistical Physics of High-Dimensional Learning – ENS Paris <i>Research Project – Advisor: Prof. Remi Monasson</i> • Designed controlled high-dimensional interpolation experiments showing how covariance-spectrum decay determines whether an autoencoder recovers smooth latent dynamics or fails on sharp, high-frequency data. • Derived with the replica method an explicit training-error expression for an asymmetric autoencoder as a function of the covariance spectrum and sample-to-dimension ratio, separating under- and over-parameterised regimes.	Sep 2025 – Mar 2026 Paris, France
Plasma Modeling for Air-Breathing Electric Propulsion – Ecole Polytechnique <i>Group research project – Advisor: Dr. Federico Petronio</i> • Developed and benchmarked a nine-species global plasma solver; mapped atmospheric composition, discharge power and losses to the thrust–drag balance, establishing a feasibility window for very low Earth orbit.	Sep 2024 – Sep 2025 Palaiseau, France
Complex Amoebas and Tropical Geometry – Ecole Polytechnique <i>Mathematical Essay – Advisor: Prof. Omid Amini</i> • Reconstructed proofs of the Hausdorff convergence of rescaled complex amoebas to tropical curves and of the algebraic–tropical zero-set correspondence using Ronkin functions and non-Archimedean valuations.	Sep 2024 – Jan 2025 Palaiseau, France

ADDITIONAL EXPERIENCE

Thales <i>Internship as Innovation & Emerging Technologies Analyst</i> • Evaluated 15+ deep-tech startups and authored an internal report mapping Japan’s quantum-computing ecosystem for strategic decision-making.	Jun 2025 – Sep 2025 Tokyo, Japan
French Navy – Frigate Aquitaine <i>Military service as Officer Cadet</i> • Held Officer-of-the-Watch responsibilities during a 12-week NATO deployment in the Baltic Sea and led a 12-sailor team.	Nov 2023 – Apr 2024 Brest, France

SKILLS

Programming: Python (NumPy, SciPy, PyTorch, Matplotlib), OCaml, Java, Git, LaTeX	
Scientific computing: Numerical continuation, ODE/DDE integration, spectral and Lyapunov analysis	
Languages: French (native), English (IELTS 8.5/9), Italian (advanced)	

LEADERSHIP & INTERESTS

Teaching: Mathematics & physics oral examiner for CPGE students	
Service: Raised €16k for education projects in Senegal; managed €40k in microcredit operations benefiting 120+ families in Guatemala	
Interests: Competitive swimming, Model UN (Best Speaker, Geneva), scouting (10+ years), chess, poker	